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1 Abstract

This report analyses engineering materials, including common ferrous and cuprous alloys, and engineered ceramics. It covers manufacturing processes, applications, and compositions. Specifically, the compositions of various ferrous metals utilised in automotive components and their properties are investigated to understand the justification for their application. Surface treatments are also evaluated. The report also outlines cuprous alloys, their composition, applications, and reasons for use in electrical applications. It discusses forming methods for cuprous alloys and major engineering ceramics, their applications, and manufacturing processes. Finally, it examines the validity of artificial intelligence-based proposals.

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2 Introduction

Engineered materials are ubiquitous, from everyday items to advanced technology. This report examines case studies of these materials in real-life contexts, with diverse examples representing their diverse applications. Ferrous alloys, commonly found in automotive components, are discussed, along with cuprous alloys used in electrical applications due to their synergistic characteristics. Engineered ceramics, with granular control over properties and composition, are also explored. The growing capabilities of artificial intelligence, however, begs the question: can engineers trust AI? This report also seeks to answer this question, ultimately with the goal of presenting a high-level overview of engineered materials in context.

3 Ferrous Metals

3.1 Types of Ferrous Metals

3.1.1 409 Stainless Steel



Figure 1: **409 Stainless Steel (Exhaust Pipes)**

(Rhino Exhaust, 2018)

Applications	Nominal Composition (S40910)	Properties & Justification for use
409 stainless steel is commonly used for automotive exhausts but is still considered a general-purpose stainless steel. Though inferior to 304 stainless, it's suitable for applications requiring heat resistance and durability, sacrificing aesthetics for performance. It's also weldable. (Atlas Steels, 2021)	$\sim 0.08\%C$, $\sim 1\%Mn$, $\sim 1\%Si$, $\sim 0.045\%Si$, $\sim 0.03\%S$, $\sim 11.1\%Cr$, $\sim 0.5\%Ni$, $\sim 0.6\%Ti$, $\sim 85.645\%Fe$ (Atlas Steels, 2021)	The extreme temperatures exhaust systems are exposed to during regular use necessitates a corrosion and heat-resistant steel. While aesthetically less desirable due to surface oxidation, the increased service life relative to its cost in comparison to more expensive examples <i>i.e.</i> 304 stainless, more than justify its use. (Torqit, 2021)

3.1.2 Cast Iron

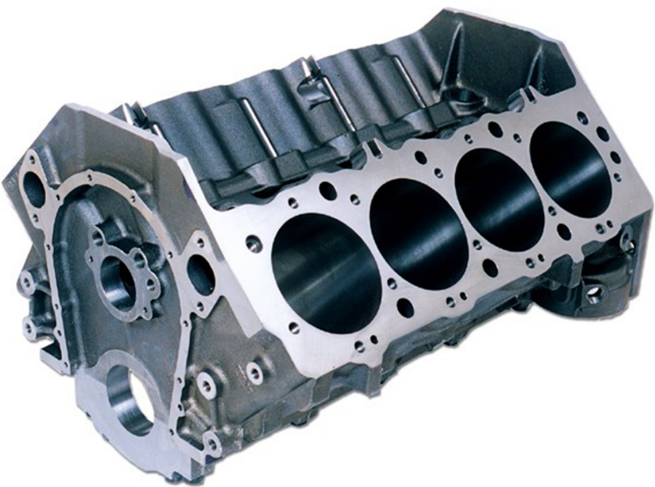


Figure 2: **Cast Iron (Engine Block)**

(VPW, n.d.)

Applications	Nominal Composition (White Cast Iron)	Properties & Justifica- tion for use
The most obvious use of cast iron parts is in engine blocks and gearbox cases, where its ease of manufacture as well as excellent durability lends itself to see usage even into the age of engineered composites. Cast iron isn't limited to gear-boxes, however, seeing extensive use in braking systems as well. (Acme Foundry, 2023)	~ 2.4%C, ~ 0.525%Mn, ~ 0.2%S, ~ 1.3%Si, ~ 0.125%P, ~ 95.45%Fe (One Stop NDT, 2021)	Cast iron's popularity is due to its mechanical properties, low cost, and ease of manufacture. Copeland (2020) highlights its excellent properties, including low cost, high compressive strength and rigidity, easy machinability, and high fluidity. These properties make cast iron a convenient material for its affordability.

3.1.3 Low-carbon Steel



Figure 3: Low-carbon (Mild) Steel

(Sukarman et al., 2023)

Applications	Nominal Composition (AISI 1010)	Properties & Justification for use
Mild steel is often the material of choice for nonstructural automotive components, such as crumple zones and impact-deforming panels. Low-carbon steel is unique in that it deforms upon impact, unlike reinforced steels that either don't deform or shatter. Mild steel is also easily recyclable, making it suitable for sustainable design. Its affordability and widespread availability make it an ideal solution for nonstructural parts.	$\sim 0.105\%C$, $\sim 0.4\%Mn$, $\sim 0.04\%P$, $\sim 0.05\%S$, $\sim 99.405\%Fe$ (AMC, 2024)	It is precisely the mechanical properties that make low-carbon steel undesirable for other applications, that crown it the <i>de facto</i> material of choice for body components. It's necessary for the crash structure to deform upon impact to reduce the energy transferred to any occupants; hardened tool steels or cast iron, both brittle and highly dense, are unsuitable for this task. The mechanical ability of low-carbon steel to deform freely on impact is invaluable in an automotive. ("Advantages of Using Mild Steel in Car Bodies", 2025)

3.1.4 High-strength-low-alloy Steel



Figure 4: High-strength-low-alloy (HSLA) Steel

(ArcelorMittal, n.d.)

Applications	Nominal Composition	Properties & Justification for use
HSLA steels regularly find use in automotive chassis, their superior strength-to-weight ratio extracting extra performance while remaining safe. (“Pros and Cons of HSLA Steel”, 2024) Specifically, bumpers, suspension systems, and structural panels are typically constructed of HSLA steel, noting that these parts are both stronger and lighter than alternatives.	Unfortunately, due to the greatly varied nature of HSLA steels, the following nominal composition is only a reference and is likely not representative of any individual example: ~ 0.25%C, ~ 2%Mn, ~ 97%Fe. The remaining weight is made up of some combination of: V, Nb, Cr, N, Mo, Ni, Cu, Zr. (Masteel, 2017)	The “exceptional strength and toughness” of HSLA steels under high-intensity applications is ideal for materials in the automotive industry. (“Pros and Cons of HSLA Steel”, 2024) Not only is HSLA steel remarkably durable, its (relative) freedom in alloying composition enables material engineers to specify properties down to a microscopic scale, incentivising targeted solutions via its excellent customisability.

3.2 Surface Treatments of Ferrous Metals

3.2.1 Nitriding of Engine Components

Steels found in engine components are exposed to extreme mechanical loads and temperatures; the crankshaft is a common example of nitrided steel, necessary for its additional surface hardness to mitigate wear while retaining the mechanical properties of the substrate. As a case-hardening process, nitriding infuses nitrogen into the surface of the steel, forming a superficial compound layer with superior hardness. (Sild, 2023) This process is best explained with a diagram.

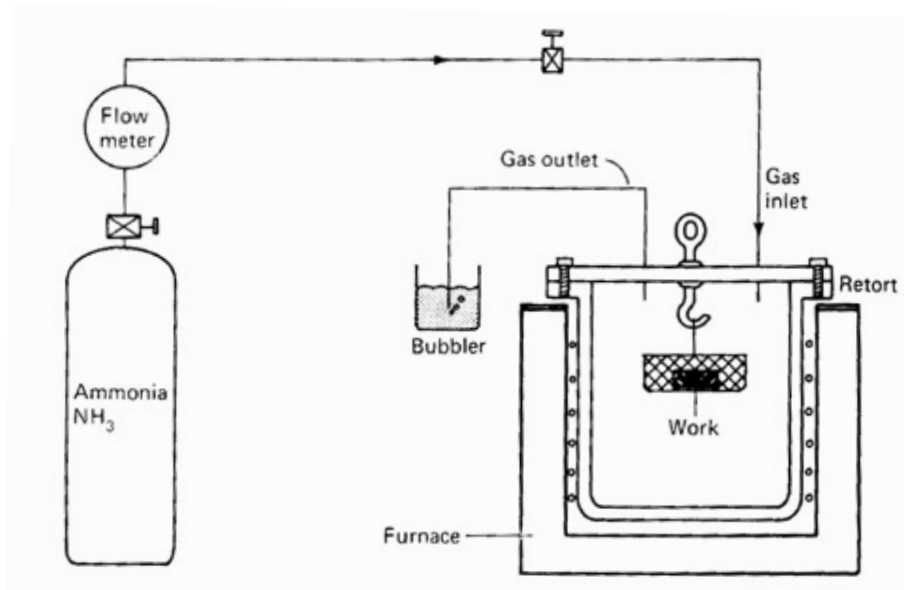


Figure 5: The gas nitriding process (“Nitriding - How it works?”, n.d.)

At a basic level, the steps of (gas) nitriding are as follows:

1. The part to be nitrided is placed in a furnace
2. The atmosphere of the furnace is replaced with anhydrous ammonia gas ($\text{NH}_{3(g)}$)
3. The temperature is increased until the ammonia decomposes into nitrogen gas ($\text{N}_{2(g)}$) and diffuses into a nitride layer chemically integrated into the steel
4. The part is held in this state until the desired nitride layer thickness is reached

The specific outcomes of nitriding include excellent surface hardness while maintaining core ductility (Sild, 2023), as well as increased chemical (corrosion) resistance, both desirable for engine components due to the frequent shock loads that they may experience. Not only does the additional chemical resistance protect from oxidation stemming from thermal cycling, but the surface hardness also reduces frictional wear; this is highly beneficial on contact components e.g. the crankshaft.

3.2.2 Electroplating of Trim Panels

While the mechanical properties of a part may be the biggest consideration to an engineer, the average customer is likely much more concerned with the aesthetic appeal of the vehicle. Electroplating is one method suitable for achieving these aesthetic outcomes, where any desired appearances can be easily bonded as a superficial coating which additionally protects the substrate. A standard setup for electroplating is shown in the following diagram.

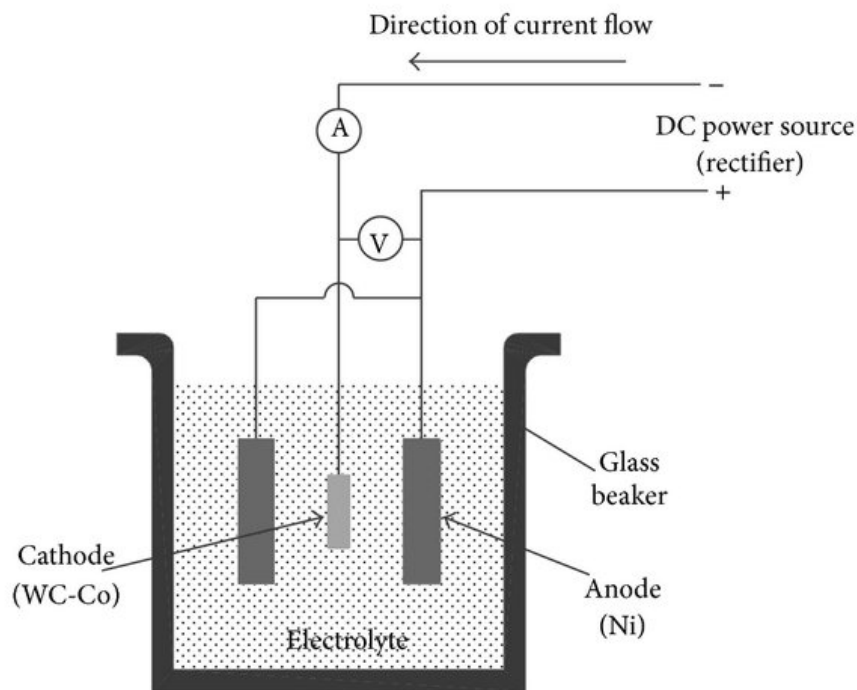


Figure 6: A standard electroplating cell (ABDELKARIM et al., 2013)

Electroplating utilises controlled electrolysis to coat the substrate with a pure layer of metal. Common options include copper, chromium, zinc, and nickel, but any metal available as an ion (solution) can be utilised. (FormLabs, n.d.) The above diagram illustrates a scalable setup, where the dissolved ions are driven to the cathode (substrate) via an applied current; the process is largely identical regardless of part size. In automotives, electroplating is often employed to give parts a metallic finish not only for aesthetics, but also protection to maintain the longevity of the components, allowing for aesthetic specifications which may be otherwise impossible or highly impractical *e.g.* utilising pure gold trim for bumpers, etc. Aesthetically pleasing visuals are a crucial element of automotive design, and electroplated parts are one such way to fulfil this need.

3.2.3 (Spray) Painting of Body Panels

Practically speaking, almost all cars in existence are (spray) painted in some capacity. Painted surfaces offer both aesthetic appeal and corrosion resistance, both valuable properties in automotive contexts. These paints may also be engineered to exact specifications, allowing for additional design flexibility. This process is usually automated, as shown below.



Figure 7: A robotic spray booth (ASA, 2024)

Some benefits of painting parts include:

- Resistance to corrosion
- Aesthetic appeal
- Protection against environmental wear

It's necessary for parts exposed to the elements in routine use to be shielded from degradation and damage, and painting provides an optimal solution that fulfils both these requirements.

3.2.4 Carburising of Gearbox Components

Similar to nitriding, carburisation is another case-hardening process utilised in similar applications. In automobiles, gearbox components are a common example of carburised steel, where durable contact surfaces are necessary, but brittle materials are unacceptable. Carburisation provides an alternative to nitriding in this case, though it is less practically effective and efficient, requiring higher temperatures but unable to achieve the same level of surface hardness.



Figure 8: A carburising furnace (AHT & McNamara, n.d.)

The process of carburising is similar to nitriding:

1. The part is held in a furnace with hydrocarbon gases and heated
2. The hydrocarbons decompose and the free carbon diffuses into the steel
3. The part is quenched to form a martensitic layer and lock in the carbon

MacKenzie (2017) outlines the widespread application of carburising in manufacturing gears; contact surfaces within gearboxes require extremely high surface hardness, but must maintain a degree of elasticity to provide a margin of safety in cases of shock loading. Carburisation fulfils this similarly to nitriding, leaving a hard case but ductile interior.

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4 Non-ferrous Metals

4.1 Copper-based metals

4.1.1 Elemental Copper

Copper as an element is second only to silver in conductivity and exhibits excellent ductility and malleability, all properties highly desirable in electrical applications.

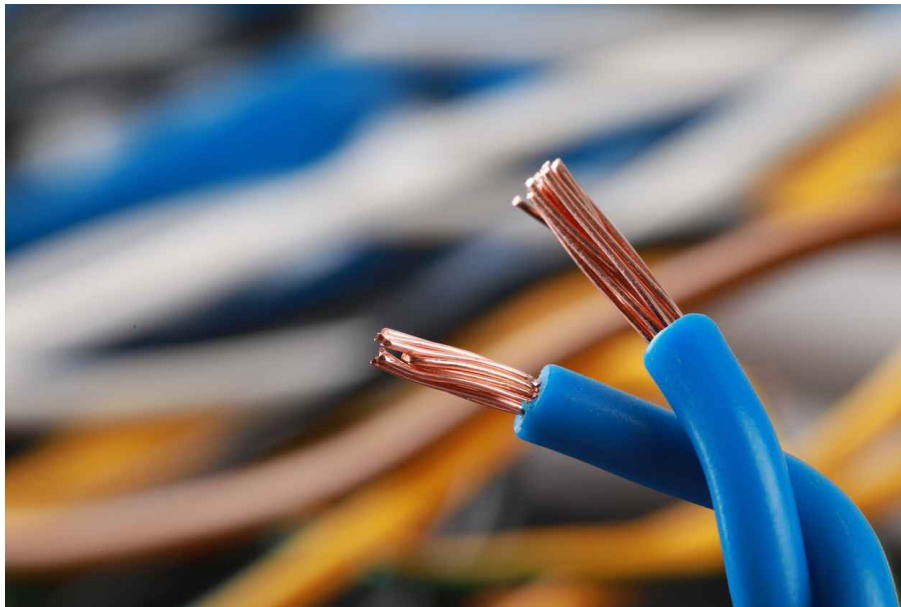


Figure 9: Some copper wires (Chesapeake Electric, 2024)

Copper isn't just found in wires, however. It finds use in various components ranging from electrical terminals to motors, but all of these take advantage of a few major properties of copper:

- Electrical & thermal conductivity
- Ductility & malleability
- Chemical resistance
- Infinite recyclability

(Copper Sustainability Partnership, 2023)

While conductivity is naturally the paramount property to consider for electrical applications, the other properties elevate copper to an invaluable engineering material. Not only does its high ductility and recyclability lend itself to easy manufacturing and use, but its chemical resistance ensures longevity once installed, withstanding even extreme industrial loads.

4.1.2 Brass

While brass measures objectively lower in thermal and electrical conductivity when compared to its parent metal, copper, it makes up this deficit through its higher strength. Similar to copper, brass is also utilised for wires and endpoint terminals, as well as switch connectors.



Figure 10: Bundled brass wire (Super Metal Industries, 2016)

A standard nominal composition for brass is as follows:

- $\sim 70\% \text{Cu}$
- $\sim 30\% \text{Zn}$

(Metal Supermarkets, 2024)

While other alloying additions may be varied to adjust the final properties, this proves a baseline of expectation; brass is also more inexpensive compared to pure copper, proving to be an important trait in today's economy. The extra hardness of the 70-30 alloy also enables easy machining of components, allowing for intricate geometries without structurally unsound parts e.g. CPU pins. An image is included below for clarity.

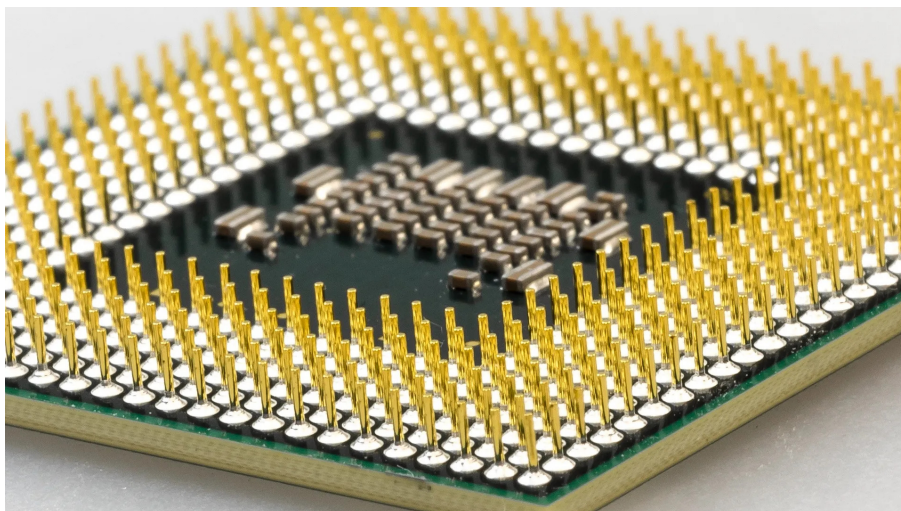


Figure 11: Brass CPU pins (SoftwareG, n.d.)

4.1.3 Chromium, Beryllium, and Zirconium Copper

While all 3 of these metals may be alloyed with copper with varied resulting properties, this report will only discuss beryllium copper for the sake of brevity, noting its comparatively excellent strength.



Figure 12: Beryllium copper (BeCu) pieces (Australian Metals, n.d.)

A typical nominal composition for beryllium copper (C17200) is as follows:

- ~ 98%Cu
- ~ 2%Be

(Australian Metals, n.d.)

Boasting strength comparable to alloy steels, beryllium copper is ideal for use in contact components such as switches and relays. In particular, while it's not utilised for wiring due to its high strength, it still finds use in connectors between components. Some specific properties and their impacts are as follows:

- Comparable strength to steels allows for use in conductive yet load bearing parts
- Moderate conductivity reduces, but doesn't eliminate use in electrical parts

(Aviva Metals, n.d.)

While the conductivity of beryllium copper is roughly one-fifth that of pure copper, it retains enough conductivity to justify use. Contact components specifically require materials capable of withstanding repeated mechanical load, making beryllium copper an ideal compromise between durability and conductivity (efficiency).

4.2 Forming Copper for Electrical Applications

4.2.1 Cold Working (drawing, extrusion, etc.)

Die Drawing The ductile nature of copper and most cuprous alloys lends itself to cold working with relative ease, with drawing and extrusion being prime examples. Illustrated below is a die for drawing wires.

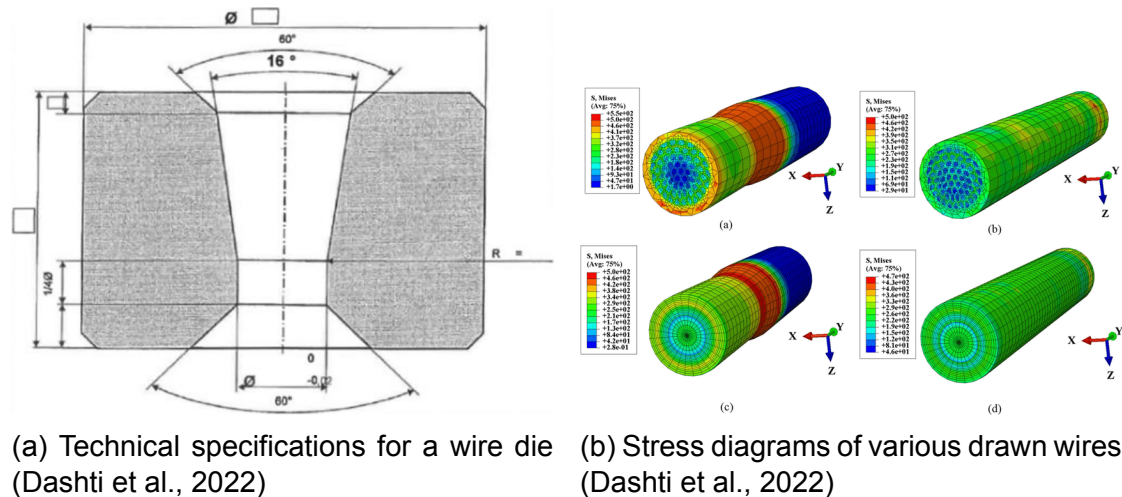


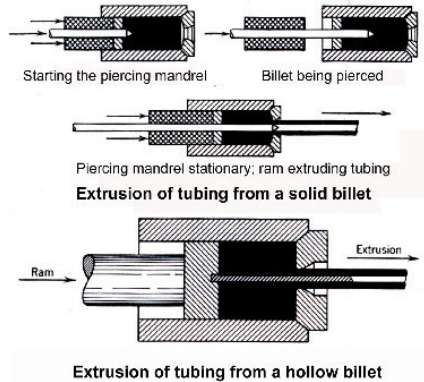
Figure 13: Various wire drawing diagrams

The process of die drawing can be summarised as follows:

1. The material pre-shaped into a suitable diameter from which to reduce
2. The wire is sequentially passed through a series of dies with decreasing diameter
3. As the wire lengthens, the diameter decreases

Notice that the wire is *drawn* through the dies as compared to extrusion, where the material is *pushed* through the dies. Wires and cables are formed via die drawing for its economic efficiency, while more complex geometries must be formed through machining or extrusion. However, most, if not all wires used in electrical applications are drawn.

Extrusion The process of extrusion is both similar and vastly different from die drawing. While the material is passed through a narrow opening (the extrusion die), it's instead pushed through with hydraulic pressure (typically) while heated above the re-crystallisation temperature.



(a) A diagram of the extrusion process (Lee, 2004)



(b) Examples of extruded parts (Puglia, 2022)

Figure 14: Diagram and example of extruded parts

While the specific part illustrated in the above figure may not match any specific electrical components, parts like busbars, connectors, and terminals are common candidates for extrusion manufacturing. Some specific benefits of extrusion include:

- Complex, continuous geometry capabilities
- Economically efficient
- Effective surface finishing

If the specification of *electrical applications* is to be taken quite literally, then the aforementioned components highlight the benefits of extrusion, allowing for more complex geometries than die drawing while maintaining part continuity. However, if this specification is taken more loosely, then examples may also include heat sinks and other components used in electrical applications, albeit not as an electrical conductor. An image is included below.

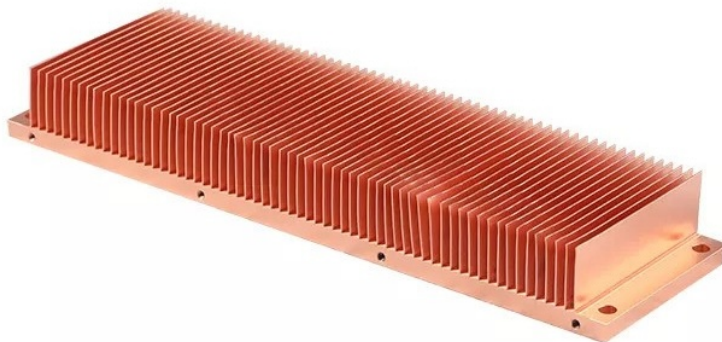


Figure 15: (Extruded) copper heat sink (Winshare Thermal, n.d.)

4.2.2 Casting

Copper may also be cast for electrical applications, allowing for the production of complex geometries that may be unrealistic to form by other means. Specifically, casting allows for the production of substantially larger parts compared to other methods without sacrificing desirable mechanical properties utilising varying casting techniques. For the sake of example, some casting techniques include:

- Sand casting
- Continuous casting
- Centrifugal casting
- Various other mould-based methods e.g.
 - Shell moulding
 - Gravity die
 - Chill casting

This report will discuss continuous and sand casting, specifically.

Continuous casting While less common in electrical applications directly, continuous casting is still an essential method in producing material "blanks" for extrusion and die drawing. The process is rather simple: molten metal is constantly passed through a thermally conductive mould that is continuously cooled to form an uninterrupted part that can then be cut to size or further refined. ("Comprehensive Guide to Copper Castings", 2025)

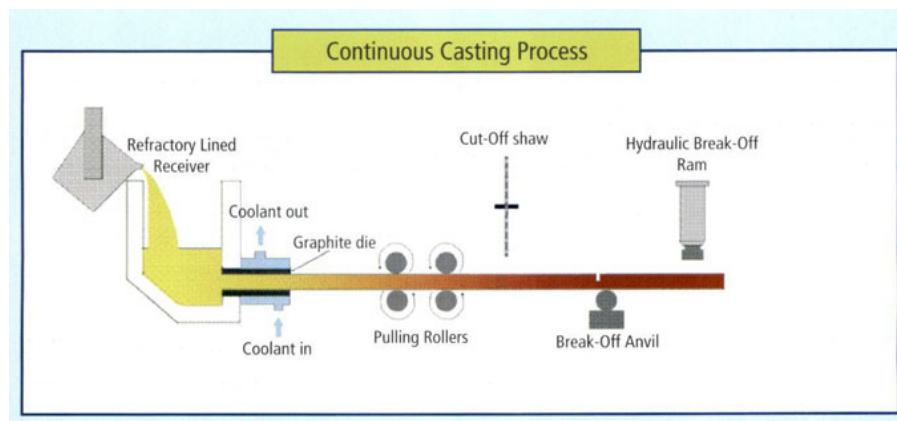


Figure 16: Continuous cast copper ("Do you want to know more about continuous casting process for copper alloys?", 2014)

Sand casting One of the age-old methods of forming, sand casting is characterised by its well-known procedure and developed toolkit for complex volumes. These parts are common in high-volume high-intensity applications such as motors and generators, as well as relay connectors where function supersedes form. Indeed, the issues of poor surface finishing or uneven cooling may be unavoidable, but the flexibility of sand casting is a justifiable compromise. (“Comprehensive Guide to Copper Castings”, 2025)



Figure 17: Sand cast bushings (Electric Materials, n.d.)

A reference procedure is as follows:

1. Reference creation
2. Mould fitting & packing
3. Inner void fills
4. Add risers and vents
5. Melt metal & pour
6. Once cooled: remove from mould and process as necessary

4.3 References

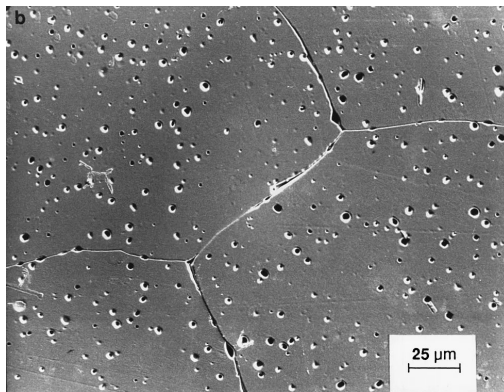
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5 Ceramics

5.1 Engineered Ceramics

5.1.1 Oxide Ceramics

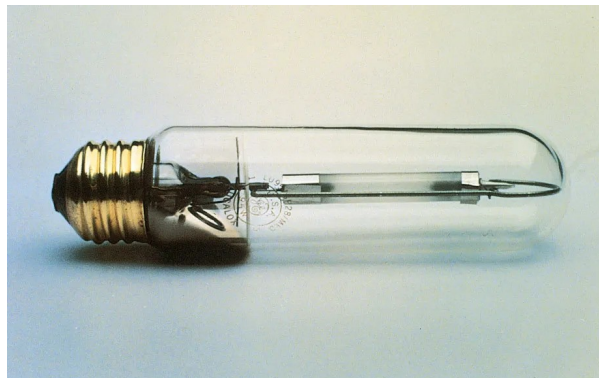
Alumina Synthetically produced, alumina (Al_2O_3) is the most common engineering ceramic and typically a first choice when considering options. Raw alumina powder may be calcinated to form ceramics with well-known properties: electrical insulation, chemical resistance, and extreme hardness. (The Editors of Encyclopaedia Britannica, 2025a; TWI, n.d.) Some images of alumina are included below.



(a) Alumina microstructure (The Editors of Encyclopaedia Britannica, 2025a)



(b) Some alumina parts (Great Ceramic, n.d.)



(c) Sodium vapour lamp with alumina enclosure (The Editors of Encyclopaedia Britannica, 2025a)

Figure 18: Alumina examples

Some applications of alumina include:

- Spark plug insulation
- Bone implants
- Abrasive tools
- Industrial refractory linings

(The Editors of Encyclopaedia Britannica, 2025a)

Zirconia Similarly to alumina, zirconia (ZrO_2) is another chemically resistant ceramic with high hardness, but differs in toughness – while alumina is brittle, zirconia instead exhibits a remarkable resistance to crack propagation and fracturing. Although zirconia is considered an engineering ceramic, it's likely better known in its *cubic* form, resembling diamonds. While the aforementioned mechanical properties and extremely high melting point (2700°C) are indeed attractive, unstabilised zirconia rapidly degrades at higher temperatures, nullifying any value it brings. Some examples are shown below:



(a) Zirconia plates



(b) Zirconia rings

Figure 19: Some zirconia examples (Great Ceramic, n.d.)

Similarly, the major applications of zirconia include:

- Dental implants
- Jewellery
- Temperature sensors
- Thermal barriers

(Zircon Industry Association, n.d.)

5.1.2 Carbide Ceramics

Silicon carbide A common trend among engineering ceramics is their extraordinary hardness; silicon carbide (SiC) is second only to zirconia out of the common engineering ceramics. (TWI, n.d.) Silicon carbide is also considered a semiconductor, particularly in IC (integrated circuit) boards and LED (light-emitting-diode) substrates. (The Editors of Encyclopaedia Britannica, 2025b) Below is an example of a SiC crystal.



Figure 20: A silicon carbide crystal (ruler for scale) (The Editors of Encyclopaedia Britannica, 2025b)

As previously mentioned, applications include:

- Abrasive tools
- Semiconductors
- Refractory linings
- Military armour

(The Editors of Encyclopaedia Britannica, 2025b; TWI, n.d.)

Tungsten carbide The industry standard for cutting tools and industrial machines, tungsten carbide (WC) is another sintered ceramic finding widespread use in a diverse range of industries. While it presents as a grey powder in its pure form, it may be sintered or cemented with carrier components to create tools. (Dymet Alloys, n.d.) Tungsten carbide is often bonded with cobalt (Co) during sintering to raise its density and toughness, but at a proportional cost to hardness. (TWI, n.d.) Below is a property table of WC alongside WC-tipped drill bits.

Tungsten carbide	
 Tungsten carbide milling bits	
Identifiers	
CAS number	12070-12-1 ✓
Properties	
Molecular formula	WC
Molar mass	195.86 g·mol ⁻¹
Appearance	Grey-black lustrous solid
Density	15.8 g·cm ⁻³ , solid
Melting point	2,870 °C, 5,198 °F (3,143 K)
Boiling point	6,000°C, 10,832 °F (6,273 K)
Solubility in water	Insoluble.
Structure	
Crystal structure	Hexagonal, hP2, space group = P6 ₃ /mmc, No. 187 ^[1]
Hazards	
EU classification	Not listed
Related compounds	
Other anions	Tungsten boride Tungsten nitride
Other cations	Molybdenum carbide Titanium carbide Silicon carbide

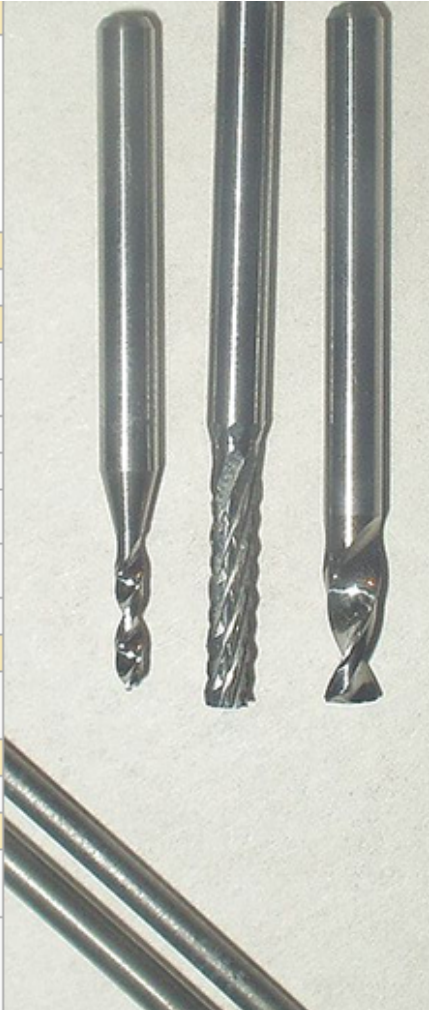


Figure 21: Tungsten carbide fact sheet & WC-tipped drill bits (Dymet Alloys, n.d.)

While tool-making is the most common and often first thought of application of WC, some other applications include:

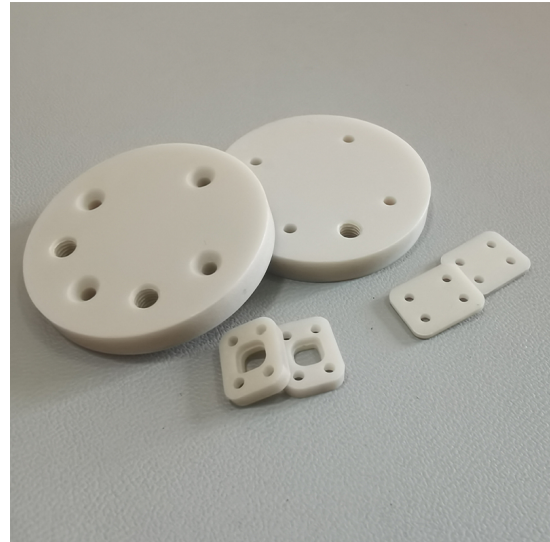
- Mining drills
- Military weapons
- Aerospace components

5.1.3 Nitride Ceramics

Aluminium nitride The exceptional thermal shock resistance and conductivity of aluminium nitride (AlN) combined with its low thermal expansion ratio grant it a unique suitability for use in aerospace and electrical engineering. Manufacturing is similar to other engineering ceramics; AlN powder is sintered with a carrier to form parts.



(a) AlN plates



(b) AlN disks and plates

Figure 22: Aluminium nitride ceramics (Great Ceramic, n.d.)

Often deemed an indispensable element of modern electronics, some applications include:

- Heat sinks and baseplates
- Inter-component insulation
- LED substrates
- Piezoelectric devices

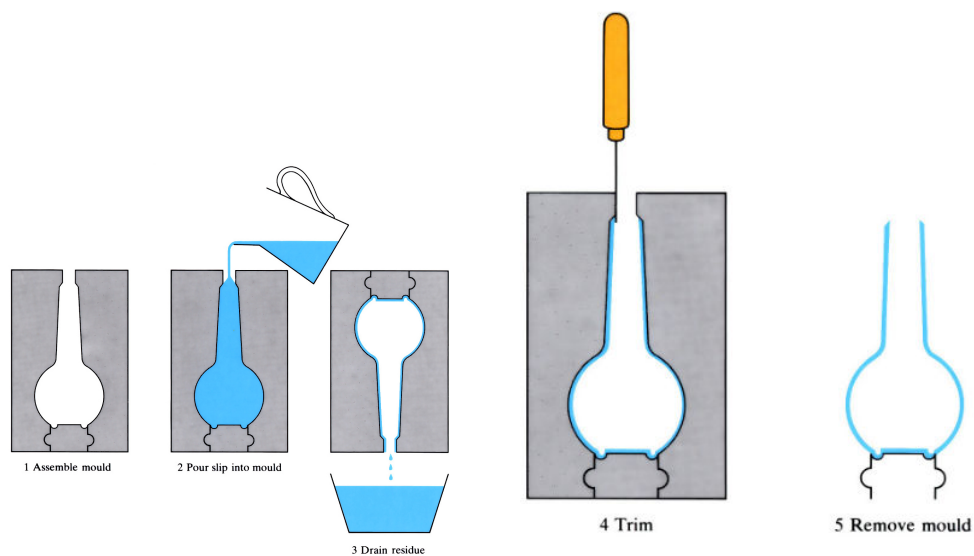
(Yang, 2024)

5.1.4 Forming Methods of Ceramics

The manufacturing processes of ceramics may be generalised into a few main stages:

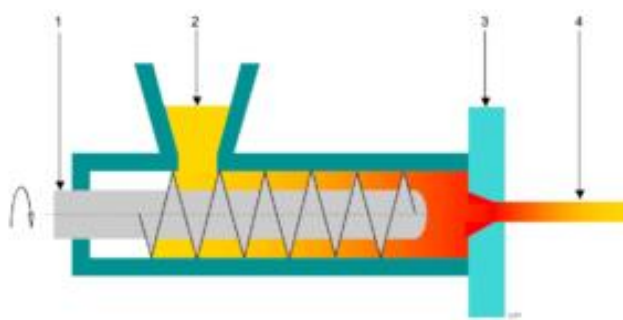
- Material preparation
- Forming (moulding)
- Sintering
- Post-processing

Within this forming stage, there are generally 3 options: slip casting, extrusion, and pressing. (“About Ceramic Manufacturing”, 2019; “Manufacturing Process of Ceramic Products: A Guide for Materials Engineers”, 2024) Diagrams of all 3 are included for clarity.

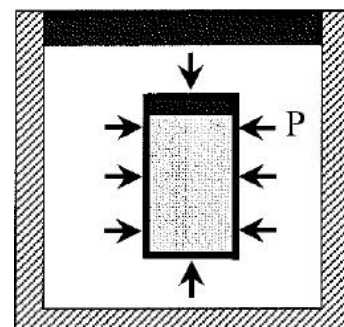


(a) Slip casting, part 1 (The Open University, 2017)

(b) Slip casting, part 2 (The Open University, 2017)



(c) Extrusion diagram (SICT, n.d.-a)



(d) Ceramic pressing diagram (SICT, n.d.-b)

Figure 23: A collection of diagrams

Slip casting In essence, this is as simple as filling a mould with liquid clay. A high level process would be as follows:

1. Prepare the mould & slip
2. Pour the slip into the mould & leave the desired length of time necessary
3. Drain the remaining slip & wait until leather-hard
4. Demould the part & process as necessary

(Simplified from Hansen, n.d.; The Open University, 2017)

With reference to **Figure 23:a, 23:b**, steps 1-3 similarly show this process. Step 4 combines **Figure 23:b** into a singular step.

Extruding Similar to metallic extrusion as discussed in **Section 4.2.1**, ceramics (and by extension, clay) can be formed in the same way. While differing slightly in the use of a screw rather than a hydraulic ram to drive the material, the process remains largely the same as compared to the aforementioned metallic process. However, a standard process is still provided:

1. Material preparation
2. Loading the extruding machine
3. Extrude the material through dies to form the desired shape
4. Once dry, trim to desired length & await further processing

The extruding machine is depicted in **Figure 23:c**.

Pressing Perhaps the simplest out of the 3 methods, pressing is as the name suggests: press the material into the mould as hard as necessary to form the desired shape. Although there are 3 types of pressing (uniaxial, isostatic, wet), the principle is identical for all:

1. Load the mould with material
2. Apply pressure until desired density is reached
3. Wait until touch-dry, then remove and post-process

Isostatic pressing is illustrated in **Figure 23:d**.

5.2 Analysis of AI

The ChatGPT response without edits:

Method for Slip Casting a Ceramic Hand Basin

1. *Prepare the Mould: Clean and assemble a two-part plaster mould, ensuring all surfaces are smooth and dry. Apply a release agent, such as soapy water, to ease demoulding.*
2. *Pour the Slip: Fill the mould with liquid clay (slip) to the brim. Allow it to sit for 15–30 minutes to form a consistent wall thickness as the plaster absorbs moisture.*
3. *Drain Excess Slip: Pour out the remaining slip, leaving a hollow shell inside the mould. Let it set until leather-hard.*
4. *Demould the Basin: Carefully remove the mould parts without damaging the cast. Smooth imperfections.*
5. *Dry and Fire: Let the basin air-dry completely before bisque firing and applying glazes for final firing.*

(OpenAI, 2025)

Perhaps the most glaring omission in the response is the preparation of the slip. The lack of this detail in itself is sufficient to show the unlikely success of this method, as it is contingent on the quality of the slip and the mould, neither of which have given technical detail. In particular, professional guides describe a *deflocculation* process that's entirely missed by ChatGPT.

As a result, consequences of product quality may include:

- Failure of the part
- Failure of the slip to dry or slip dries too quickly
- Poor aesthetics

Hansen (n.d.) mentions the ratio of slip components as the following, noting that this slip neither formed a film nor settled overnight, indicating correct deflocculation:

- 10000g porcelain powder
- 8.8 litres water
- 44g *Darvan* deflocculant

Furthermore, Hansen (n.d.) also stresses the importance of proper slip viscosity, noting that 100ml of slip should drain from a 4.25 mm hole in approximately 40 seconds.



(a) Correctly deflocculated slip



(b) Viscosity measuring cup

Figure 24: Images from a guide by Hansen (n.d.)



(a) Incorrect deflocculation



(b) Poor results with flat moulds

Figure 25: Examples of mistakes from a guide by Hansen (n.d.)

Hence, the success of this product is entirely contingent on the quality of the slip and the mould; ChatGPT's failure to provide this technical information results in a determination of poor quality.

Hence, the revised response:

Method for Slip Casting a Ceramic Hand Basin

1. **Prepare slip with a 1.8:1 ratio of porcelain powder to water**
2. **Add deflocculant until the viscosity is such that 100ml of slip should drain from a 4.25 mm hole in approximately 40 seconds (approximately 1g of deflocculant per 100ml of water)**
3. *Prepare the Mould: Clean and assemble a two-part plaster mould, ensuring all surfaces are smooth and dry. Apply a release agent, such as soapy water, to ease demoulding.*
4. **Test slip drying time and determine the desired length of time to leave the slip in the mould**
5. *Pour the Slip: Fill the mould with liquid clay (slip) to the brim. Allow it to sit for **the length of time previously determined** to form a consistent wall thickness as the plaster absorbs moisture.*
6. *Drain Excess Slip: Pour out the remaining slip, leaving a hollow shell inside the mould. Let it set until leather-hard; **ensure this is not left for longer than necessary as cracking may occur as seen in Figure25:d.***
7. *Demould the Basin: Carefully remove the mould parts without damaging the cast. Smooth imperfections.*
8. *Dry and Fire: Let the basin air-dry completely before bisque firing and applying glazes for final firing.*

As previously presented, **Figure 23:a and 23:b** are excellent graphical examples of the procedure.

5.3 References

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6 Conclusion

This report analyses engineering materials, including ferrous and cuprous alloys, and engineered ceramics. Ferrous metals like 409 stainless steel, cast iron, low-carbon steel, and high-strength-low-alloy steel are crucial in automotive components. Surface treatments enhance their performance and longevity. Cuprous alloys, including elemental copper, brass, and specialised alloys, are vital in electrical systems. Engineered ceramics, such as oxide, carbide, and nitride ceramics, showcase their significant applications and complex forming methods. The report questions the reliability of AI in engineering, noting its potential efficiencies but emphasising the need for critical evaluation. Overall, the diverse examples and case studies underscore the vast applications and critical importance of engineered materials across industries. Staying informed about material science developments is essential for engineering professionals.